Development of Closed-loop Speed and Heater Control System in Battery Production Line

A Mid Semester Project Report

submitted by

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in partial fulfilment of requirements
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BACHELOR OF TECHNOLOGY



Department of Electronics and Communication Engineering
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MARCH 2025

DECLARATION OF ORIGINALITY

I, **PATAN IRFANKHAN**, with Roll No: **EC21B1056** hereby declare that the material presented in the Project Report titled **Development of Closed-loop Speed and Heater Control System in Battery Production Line** represents original work carried out by me in the **Department of Electronics and Communication Engineering** at the Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram.

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- I have understood that the work may be screened for any form of academic misconduct.

Patan Irfankhan

Place: Chennai

Date: 07.03.2025

CERTIFICATE

This is to certify that the report titled **Development of Closed-loop Speed and Heater Control System in Battery Production Line**, submitted by **Patan Irfankhan (EC21B1056)**, to the Indian Institute of Information Technology, Design and Manufacturing Kancheepuram, for the award of the degree of **BACHELOR OF TECHNOLOGY** is a bona fide record of the work done by him/her under my supervision. The contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

Slot-die coating is a widely used technique for the precise and uniform deposition of thin films in manufacturing industries such as electronics, batteries, and biomedical devices. However, maintaining consistent coating quality depends on two key factors: stable conveyor speed and precise temperature control during drying. Inconsistent drying temperatures can affect solvent evaporation, film adhesion, and overall coating performance, while even small variations in motor speed can result in uneven film thickness and imperfections. To address these issues, a closed-loop control system is implemented using an Arduino Uno harnessing a Proportional-Integral (PI) controller. An infrared (IR) sensor measures the stepper motor's speed in real time, and the controller adjusts the pulse delay dynamically to maintain the desired speed. This approach ensures stable motor operation, minimizes speed fluctuations.

To ensure a consistent heat distribution during the drying process, we employ thermocouples, solid-state relays (SSRs), infrared heating elements and a PID temperature controller.

By ensuring both precise speed control and stable temperature regulation, this system minimizes coating inconsistencies and enhances film quality. This method improves the scalability, dependability, and suitability of the slot-die coating process for high-precision applications.

KEYWORDS: Slot-die coating; Proportional-Integral(PI) controller; speed control ; gradient temperature; defect free coat.

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ABBREVIATIONS

EV	Electric vehicle
RPM	Rotations per minute
IR	Infrared
PI	Proportional-Integral
PID	Proportional-Integral-Derivative
SSR	Solid state relay
SPR	Steps per revolution

NOTATIONS

m/min	Meter per minute
ms	Milli second
v	Speed of the conveyor
K_p	Proportional gain
K_i	Integral gain
K_d	Derivative gain

CHAPTER 1

Introduction

1.1 Introduction

Thin film deposition is essential to the development of several industries in the modern world, including electronics, energy storage, healthcare, and aerospace. Thin films offer high-performance, lightweight, and energy-efficient materials that are being demanded as technology develops.One of the most important applications of thin films is battery technology.

Battery performance has become an issue of major concern with the growing demand for electric vehicles (EVs) and renewable energy. For next-generation energy storage solutions, thin film coatings on electrodes, separators, and solid electrolytes are crucial because they greatly improve battery efficiency, durability, and safety.Methods including Dip coating,Spin coating, Spray coating and Slot-die coating are commonly used in thin film deposition.

Slot-die coating has special properties because of its higher precision, better accuracy, efficient use of materials, and the scalability for large-scale battery manufacturing lines as presented in table 1.1 and its schematic diagram is shown in figure 1.1.

Property	**Slot-die coating**	**Spin coating**	**Spray coating**	**Dip coating**
Pre-metered material control	✓	✗	✗	✗
Large-Area scalability	✓	✗	✓	✓
High uniformity & Thickness control	✓	✓ (small areas)	✗	✗
Multi-layer coating feasibility	✓	✗	✗	✗
Low material waste	✓	✗ (High waste)	✗ (High waste)	✗ (High waste)
Handling of High-Viscosity materials	✓	✗	✗	✓
Clean & Contaminant-Free process	✓	✓	✗ (Overspray)	✗ (Liquid contamination)

Table 1.1: Comparison of slot-die coating with other coating techniques.

1.1.1 Slot-die coating

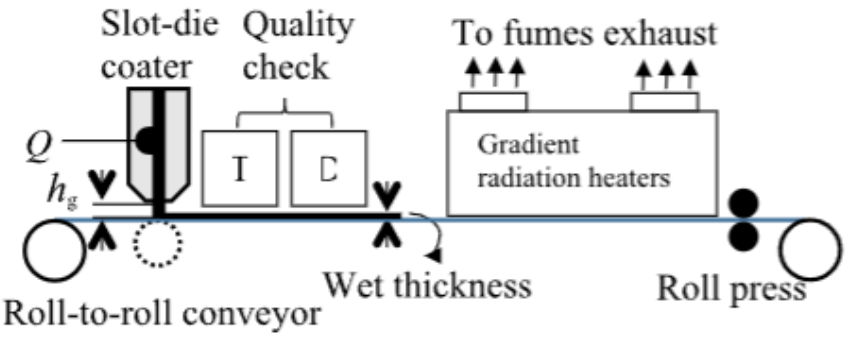


Figure 1.1: The schematic diagram of battery in-line manufacturing using slot-die coating.

The slot-die (a specialized tool designed for controlled material flow) coating is a precise manufacturing process that is used to deposit thin, uniform layers of material onto a substrate. The process involves dispensing a liquid material, such as ink, slurry, or solutions through a narrow slot in the die. This liquid is fed to a motor-driven conveyor system to create a smooth and consistent film over the substrate. The coated material undergoes a drying or evaporation process to solidify the film, ensuring its adhesiveness over the substrate to achieve the required mechanical and functional properties.

1.2 Motivation

The quality of the slot-die coating is influenced by the following aspects:

- Process parameters
- Material properties
- Equipment design

1.2.1 Process parameters

The key process parameters that influence the slot-die coating process include the coating gap (the distance between the tip of the slot-die and the substrate), the flow rate, the

substrate speed at the conveyor, and the drying process temperature. Maintaining a stable motor speed that drives the substrate is crucial to avoid inconsistencies and defects, as fluctuations can affect the uniformity and thickness of the deposited film. In a similar way, consistent solvent evaporation, film adhesion, and overall coating quality depend on accurate temperature management. Defects including uneven film thickness, cracking, and poor adhesion can arise from irregular evaporation rates caused by variations in drying temperature.

1.2.2 Material properties

The properties, such as viscosity and surface tension, can affect the spread and adhesion of the liquid to the substrate.

1.2.3 Equipment design

The design of the die and substrate plays significant role in achieving consistent coating. External factors such as temperature, humidity, and vibrations can further affect the process. Specifically, inconsistent motor speed can lead to uneven deposition causing defects and can affect overall coating quality.

- Even with a consistent flow rate, viscosity, and surface tension, achieving a uniform thin-film wet coating remains challenging due to variations in web tension and web speed (Wet coating).
- In addition, it is difficult to obtain a thin-film without defects after post-drying due to inadequate temperature control from heaters (Drying stage).
- It is observed that employing open-loop systems decreases the web speed and rapidly increases the heater temperature with increase in load. Hence, there was no room for achieving desired performance and it is necessary to employ closed-loop control systems in production lines.

1.3 Problem statement

Conveyor systems are essential for moving materials in industrial and research settings, particularly in operations involving precise manufacturing and coating. Maintaining a steady temperature gradient in a heating chamber while guaranteeing accurate conveyor speed control is a difficult task. While inappropriate temperature management impacts the quality and characteristics of materials, variations in speed might result in irregular processing.

The aim of this project is :

- To develop a closed-loop system to control the speed of the conveyor/web with an aim of achieving a stable operation and coating window under different load conditions (assuming controlled flow rate, viscosity and surface tension).
- To fine tune the Proportion-Integral-Derivative (PID) parameters of the temperature controller (Drying stage).
- Also, to maintain a gradient temperature in the drying process for a defect-free post drying coat.

1.4 Literature Review

Conveyor motor control systems commonly use DC gear motors and motor driver modules for bidirectional movement. IR, laser, or LDR sensors are frequently used in product counting processes because of their affordability and ease of use. Due to their low cost and simplicity of operation, these systems are frequently used in small-scale industrial settings [1]. However, in situations when high accuracy is required, the lack of feedback systems restricts precise motor control, and simple sensors are vulnerable to interference from ambient light, which lowers dependability. PID controllers play a key role in the functioning of conveyor belts powered by DC motors, especially when it comes to maintaining stability during changes in load. Recent simulations using MATLAB Simulink have shown how important it is to properly tune PID parameters. Research indicates that when these parameters are adjusted correctly, they can achieve low overshoot under 6% and a fast settling time of less than 1 second, with no steady-state error. These optimizations not only improve how quickly the system responds to changes but also ensure steady performance, making conveyor systems operate more

efficiently [2]. Overall, these findings highlight the importance of optimizing PID controllers in industrial settings, reinforcing their vital contribution to the reliability and effectiveness of conveyor belt systems under varying conditions. Traditional single-motor belt conveyors often struggle with problems like too much tension, inconsistent speeds, and overall inefficiency. A new solution involves using permanent magnet motors in a distributed system, which tackles these issues by employing a multi-motor ring coupling control strategy combined with double sliding-mode direct torque control.Simulations run in MATLAB/Simulink show promising results: this method helps to lower peak tension, improve speed stability, and make the system more reliable even with changing loads [3]. Although it offers benefits like better energy efficiency, easier tension management, and the ability to scale up, this system comes with increased complexity, higher costs, and greater maintenance needs. Nonetheless, distributed motor-driven conveyors are a more reliable and responsive option, making them ideal for large-scale material handling tasks.

Wireless control systems for DC motor-driven conveyor belts have emerged as a significant advancement in the realm of industrial automation, primarily by enhancing flexibility and efficiency. These systems utilize RF communication, facilitating remote operation of motor speed and direction, which positions them as suitable candidates for automated monitoring applications. Central to this process is a microcontroller that adeptly processes wireless inputs, adjusting motor parameters in real time. This capability ensures a responsive operation characterized by minimal latency.However, while the effectiveness of such remote control systems is acknowledged, it is crucial to note the absence of feedback-based error minimization techniques, such as Proportional-Integral-Derivative (PID) or Proportional-Integral (PI) control. The integration of these techniques is vital for applications requiring high precision [4]. Furthermore, existing research predominantly focuses on DC motors, suggesting a potential oversight in exploring alternative motor types like stepper motors, which may offer enhanced positional accuracy. The exploration of these factors is essential for advancing the field of industrial automation and optimizing system performance .

IoT-based systems provide a way to monitor and control the speed of conveyor lines in real-time, making industrial automation more efficient and affordable. These

systems allow for remote monitoring through tools like ThingSpeak, which helps in visualizing and managing data effectively. Some of the main benefits include the ability to customize designs for future upgrades and receiving real-time alerts. This ensures that any operational changes can be addressed promptly[5].

This study primarily focuses on addressing the challenges associated with motor speed control and gradient temperature in slot-die coating systems, aiming to ensure stable and uniform deposition of thin films for high-quality applications.

CHAPTER 2

METHODOLOGY

2.1 Porposed methodology of conveyor system

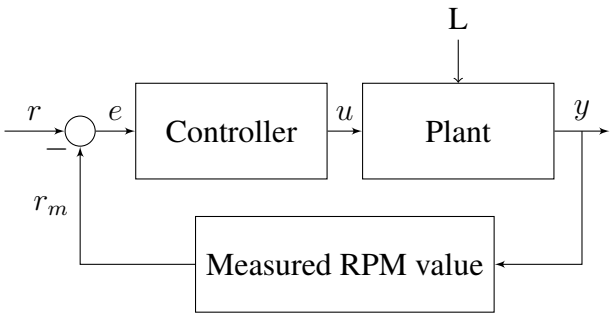


Figure 2.1: Closed-loop control system for conveyor speed control.

where :

- r = Target RPM
- r_m = Measured RPM
- e = Target RPM - Measured RPM
- u = Control signal
- L = External load
- y = Output

2.1.1 Initialization

1. Defining the Target Belt speed
   - We set the target Belt speed needed in the microcontroller.

2. Configuring the input and output pins in microcontroller
   - Set the output pins for the direction and step pulse of the stepper motor.
   - Set the input pin for the IR sensor to take the measured RPM as the feedback.

3. Working of the control system
   - The target RPM of the stepper motor is calculated using the given formula by the microcontroller:

$$v = 2\pi R \times \text{RPM} \tag{2.1}$$

   where:

   - v = Belt speed (m/min)
   - R = Roller radius (m)
   - RPM = Rotations per minute of the roller

4. Flow process
   - The microcontroller calculates the Time period for the required RPM and generates the step pulse by toggling between **HIGH** and **LOW** for a period of "T/2" each.

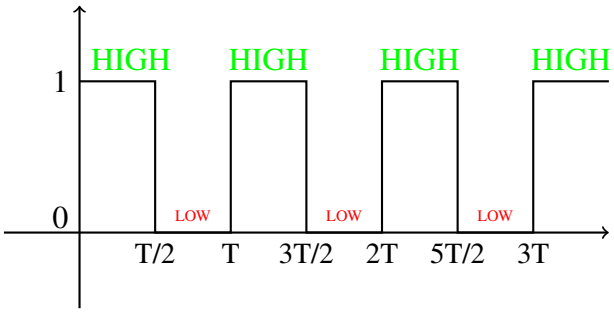


Figure 2.2: Step pulse waveform.

```
digitalWrite(steppin, HIGH);
delayMicroseconds(delayTime); // T/2
digitalWrite(steppin, LOW);
delayMicroseconds(delayTime); // T/2
```

   The generated step pulse is fed to a stepper motor driver (TB6600 in this case) through which the stepper motor rotates in a specific direction and specific speed.

5. Feedback from the IR sensor
   - An IR sensor is used to measure the RPM of the motor by counting the number of times the notch of stepper motor interrupts the IR sensor.

$$\text{RPM} = \frac{\text{revcount} \times 60000}{\text{time interval [in ms]}} \tag{2.2}$$

   This RPM is fed back to the microcontroller and Time period is calculated from the RPM, Time period relation of the stepper motor and error is calculated.

$$\text{error} = \text{desiredTimeperiod} - \text{currentTimeperiod}; \tag{2.3}$$

6. PI algorithm implementation
   - The PI controller calculates an output value based on the error between a desired set-point and a measured value. The output is a combination of a proportional term and an integral term, defined by the following equations:

$$\text{Proportional Term} = K_p \cdot \text{error} \tag{2.4}$$

$$\text{Integral Term} = K_i \cdot \int \text{e}(t)\, dt \tag{2.5}$$

Where:

- K_p is the proportional gain.
- K_i is the integral gain.
- error is the difference between the setpoint and the measured value.
- $\int \text{e}(t)\, dt$ represents the integral of the error over time.

The overall PI controller output is the sum of these two terms:

$$\text{Output} = \text{Proportional Term} + \text{Integral Term}; \tag{2.6}$$

$$\text{Output} = K_p \cdot \text{error} + K_i \cdot \int \text{e}(t)\, dt; \tag{2.7}$$

In a discrete implementation (like on a microcontroller), the integral term is approximated using a summation:

$$\text{Integral Term}[n] = \text{Integral Term}[n-1] + K_i \cdot \text{e}[n] \cdot \Delta t \tag{2.8}$$

Where:

- n is the current time step.
- Δt is the time difference between samples.

2.2 Proposed methodology for gradient temperature

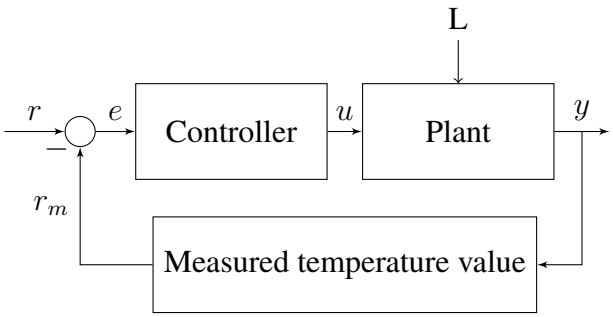


Figure 2.3: Closed-loop control system for temperature control.

where :

- r = Target temperature
- r_m = Measured temperature
- e = Target temperature - Measured temperature
- u = Control signal
- L = External load
- y = Output

The PID (Proportional-Integral-Derivative) temperature control system is implemented to ensure precise and stable temperature regulation in the drying process of the slot-die coating system. This method ensures that the temperature stays close to the desired value.

2.2.1 System components

The PID temperature control system consists of the following components:

- Infrared heating elements
- Solid-State Relay (SSR): Acts as a switching device to control the power supply to the heating elements.
- Thermocouple : Measures the temperature in real time inside the drying chamber.
- PID Controller (PT344A-K): Processes the temperature data and adjusts the heating power accordingly.

2.2.2 Working principle

The temperature control system follows a closed-loop feedback mechanism where the PID controller continuously monitors and adjusts the heating process. The steps involved are:

1. Temperature measurement

- The thermocouple senses the real-time temperature inside the heating chamber.
- The measured value is sent to the temperature controller for processing.

2. Error calculation

- The PID controller computes the error (E) between the set temperature (T_{target}) and the measured temperature (T_{measured}).
- The error is calculated as:

$$E = T_{\text{target}} - T_{\text{measured}} \tag{2.9}$$

3. PID computation

The controller processes the error using the standard PID equation:

$$u(t) = K_p E(t) + K_i \int E(t)dt + K_d \frac{dE(t)}{dt} \tag{2.10}$$

where:

- K_p is the Proportional gain (corrects based on current error).
- K_i is the Integral gain (corrects based on past accumulated errors).
- K_d is the Derivative gain (predicts future errors).

4. Control output to heating system

- The computed control signal modulates the SSR (Solid State Relay) to regulate the power supplied to the heating elements.

5. Real-time feedback and adjustment

- The system continuously monitors and updates the heating power based on temperature variations in real time.
- The PID parameters should be tuned to prevent overshooting, temperature fluctuations, and unstable drying conditions.

2.2.3 Implementation and Tuning

To optimize performance, the PID controller is tuned using the trial-and-error method or by using the auto-tuning option.

This methodology ensures reliable and efficient temperature control for defect free or crack free coating.

CHAPTER 3

EXPERIMENTAL SETUP

3.1 HARDWARE IMPLEMENTATION

3.1.1 Experimental setup

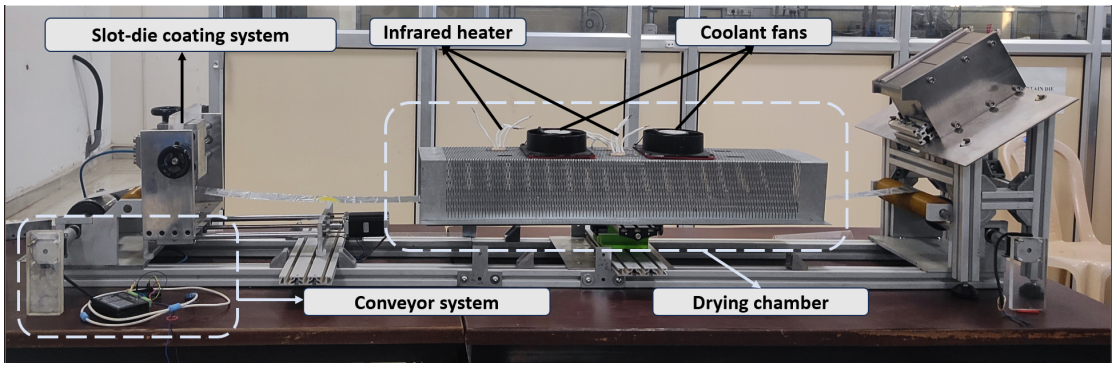


Figure 3.1: Developed battery in-line manufacturing using slot-die coating.

3.1.2 Conveyor system and motion components

- **NEMA 17 Stepper motor**
- Stepper motors offer a precise control over the speed, using this motor would be a great help in controlling the conveyor.

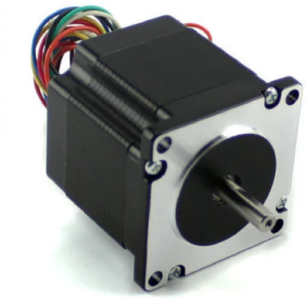


Figure 3.2: NEMA 17 stepper motor.

- **TB6600 servo drive**
- It regulate current and step pulses given to the stepper motor.
- It supports high current output (up to 2A per phase), making it suitable for driving your NEMA 17 stepper motor, which requires sufficient power for precise and reliable movement in the conveyor system.
- It also has microstepping option through which we can enhance the smoothness and accuracy of the motor movement which reduces vibration ensuring a stable motor speed.



Figure 3.3: TB6600 Stepper motor driver.

- Power supply: 24V power source for stable operation.
- Conveyor mechanism:
  - Belt-driven system mounted on two rollers.
  - Roller radius: 4 cm.
- **Infrared sensor**
- Detects roller rotation and provides real time RPM as feedback to the microcontroller.

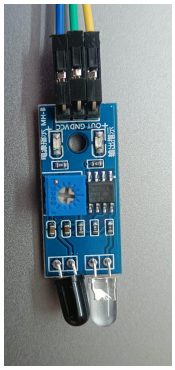


Figure 3.4: IR sensor.

- **Arduino micro controller**
- It can be programmed for automation and real-time adjustments.



Figure 3.5: Arduino Uno.

3.2 Circuit diagram

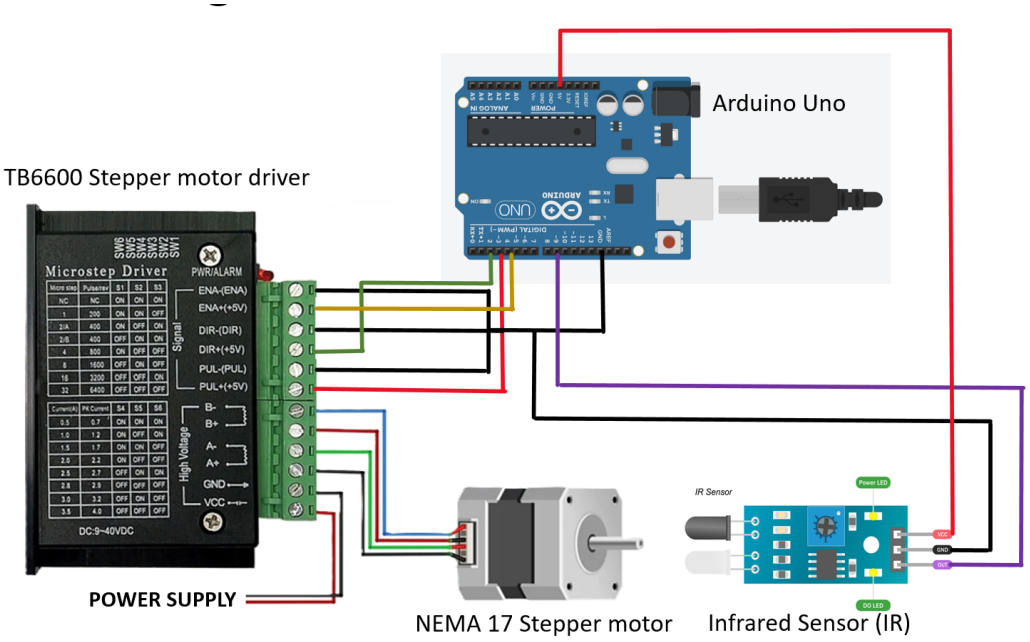


Figure 3.6: Circuit diagram of the motor control system.

TB6600 driver pin	Arduino pin
ENA+	5V
ENA-	GND
DIR+	D9 (Direction pin)
DIR-	GND
PUL+	D3 (Step pin)
PUL-	GND

Table 3.1: TB6600 stepper motor driver to Arduino connections.

IR Sensor pin	Arduino pin
VCC	5V
GND	GND
OUT	D2

Table 3.2: IR sensor to Arduino connections.

3.2.1 Heating chamber components

- **PT344A-K PID Temperature controller**.
- Maintains precise temperature control in the heating chamber. Uses a feedback loop with a thermocouple to regulate heating elements.



Figure 3.7: PT344A-K Temperature controller by sub-zero.

- **Infrared Ceramic heating element**.
- It provides uniform heating for the conveyor process.Efficiently transfer heat to materials without direct contact.
- Works well with SSR-based control for precise temperature regulation.

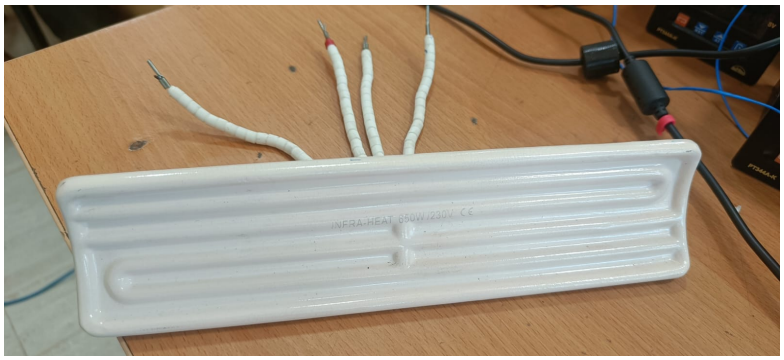


Figure 3.8: Infrared Heating elements.

- **Solid state relay for the switching.**
- It can controls the power supply to the heating elements based on temperature feedback.
- Provides fast switching and reduces wear and tear compared to mechanical relays.

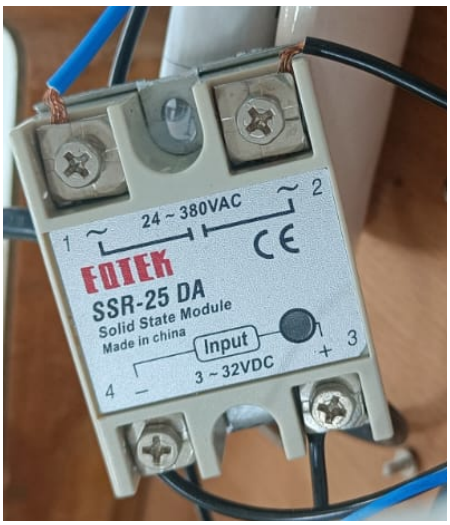


Figure 3.9: Solid state relay SSR-25 DA.

3.3 Circuit diagram

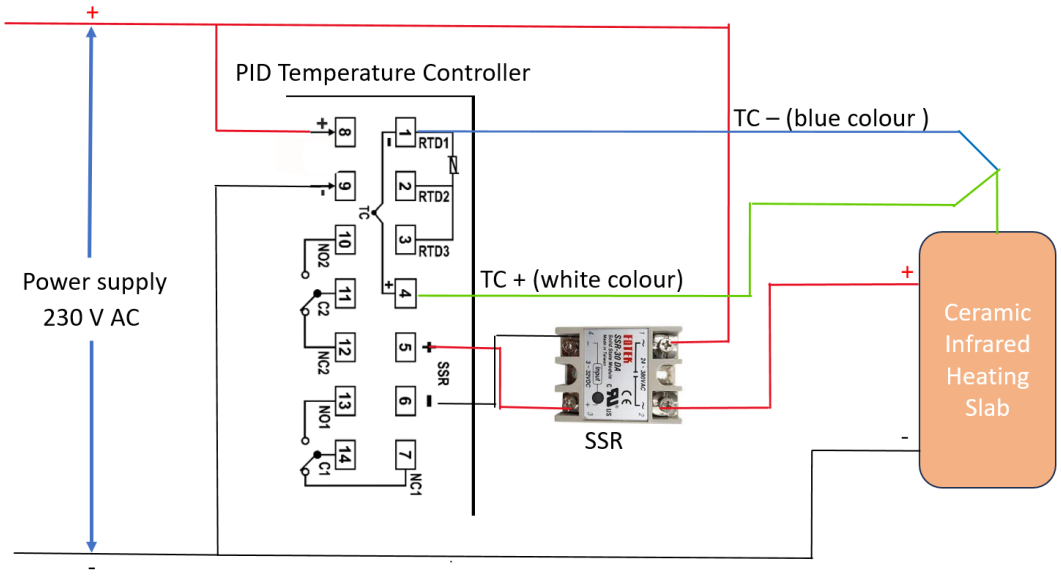


Figure 3.10: Connection diagram of the temperature control system.

Component	Connection details
Power Supply (230V AC)	Connected to PID controller (Terminal 8 (P) and 9 (N)
TC+ (White) TC- (Blue)	Connected to TC+ on PID controller (Terminal 4) Connected to TC- on PID controller (Terminal 1)
SSR Input + SSR Input -	Connected to PID SSR Output + (Terminal 5) Connected to PID SSR Output - (Terminal 6)
SSR Output + SSR Output -	Connected to heating slab + Connected to heating slab -

Table 3.3: Connections for PID temperature controller system.

CHAPTER 4

RESULTS OF THE SPEED CONTROL SYSTEM

4.1 Real time RPM plots

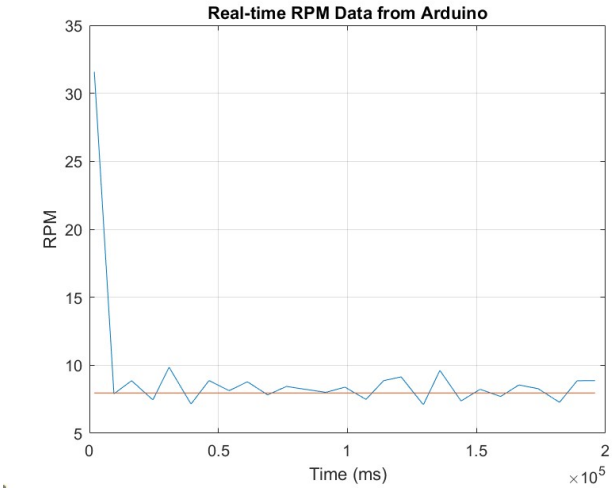


Figure 4.1: Real time RPM vs time plot for 2 m/min desired speed under no load condition.

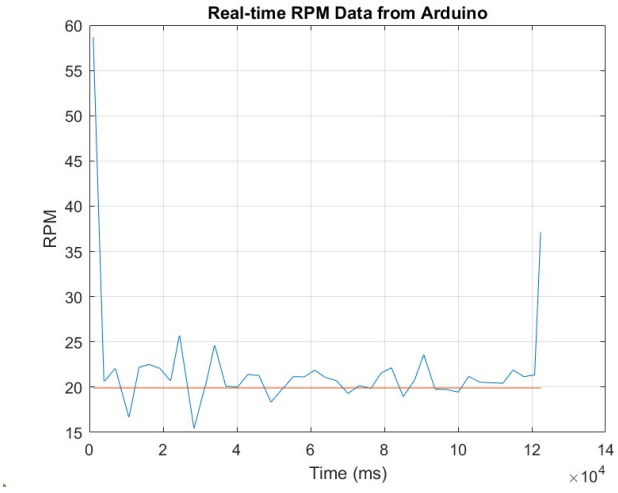


Figure 4.2: Real time RPM vs time plot for 5 m/min desired speed under no load condition.

- Obtained a stable operation of the conveyor for speeds of 2 m/min and 5 m/min under no load condition.
- The systems performance is better for low speeds.

4.2 Temperature controller results

- Tuned gain values of the temperature controller
- K_p = 43 ; K_i = 68 ; K_d = 17.



Figure 4.3: Obtained gain values (Left) Proportional-K_p, (middle) Integral-K_i and (right) Derivative-K_d.

- Constant temperature operation is observed within 15 s for desired temperatures of 50°C and 60°C.

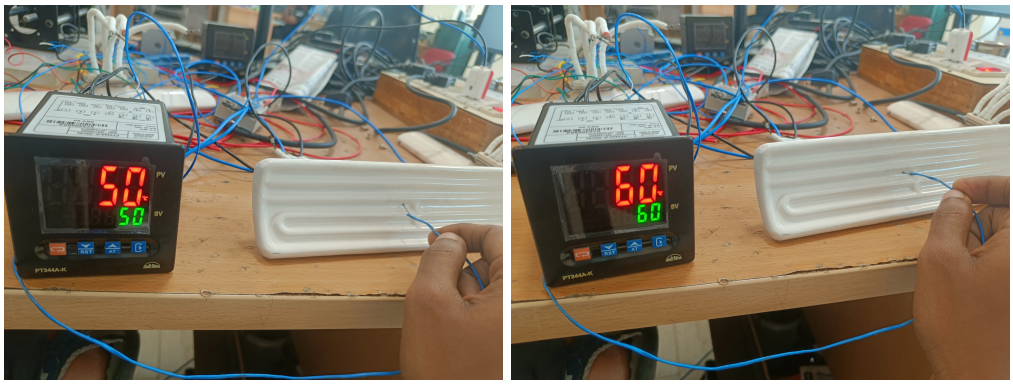


Figure 4.4: Output temperature for set points of (Left) 50 °C and (Right) 60 °C.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

In the present work, a closed-loop control system is successfully implemented to regulate both the temperature in the drying process and the speed of the stepper motor driving the conveyor system in a slot-die coating setup. The PID temperature controller achieved a stable temperature operation within 15 s by tuning the gain values of PID parameters. The gain values are K_p = 43 ; K_i = 68 ; K_d = 17. Furthermore, the speed control of the stepper motor is carried out under no-load conditions using an infrared sensor and an Arduino controller, which maintains a consistent conveyor speed. The experimental setup is tested for conveyor speeds of 2m/min and 5 m/min. The effective control of these parameters significantly improves the uniformity, reliability, and quality of the coated thin films, making the system suitable for attaining precise battery manufacturing lines.

5.2 Future scope

- By employing a closed-loop system to maintain stable web speed with different loads on the conveyor.
- Using two motors (one to each roller) at the extreme ends of the setup to maintain enough web tension. This approach reduces the burden on one motor alone for operation.
- To increase the precision of RPM measurements, IR sensors will replace with high precision encoders.
- To ensure optimized drying conditions for different materials and to enhance the quality of thin films, gradient temperatures will be employed.
- To add a syringe pump mechanism to the slot-die system for a continuous and controllable flow of slurry maintaining the same speed as the conveyor to obtain a stable coating window.
- Integration of syringe pump mechanism, gradient temperature and speed control of the conveyor systems to coat the battery electrode material.
- Finally, testing and optimizing the coating system for various substrate and battery materials.

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